



MATEMATICI 2

Curs 13

Prop.: V = sp. vect. finit, real, înzestrat cu prod. scalar

$f : V \to V$ apl. liniară

f^* = adjuncta lui f

atunci pt. $\forall$ bază ortonormală $B = \{e_1, e_2, \dots, e_n\}$ avem

$$(M_f^B)^T = M_{f^*}^B$$

Dem.:

$B = \{e_1, \dots, e_n\}$

$f(e_j) = \sum_{l=1}^{n} a_{lj} e_l$

$f^*(e_i) = \sum_{q=1}^{n} d_{ij} e_q$

$d_{ij} = d_{qi}$

$A = M_f^B$

$A = M_{f^*}^B$

$$\langle f(e_j), e_i \rangle = \langle e_j, f^*(e_i) \rangle =$$

$$= \langle \sum_{l=1}^{n} a_{lj} e_l, e_i \rangle = \langle e_j, \sum_{q=1}^{n} d_{qi} e_q \rangle$$

$$\sum_{l=1}^{n} a_{lj} \langle e_l, e_i \rangle = \sum_{q=1}^{n} d_{qi} \langle e_j, e_q \rangle$$

$$a_{ij} = d_{ji} \Rightarrow D = A^T = (M_f^B)^T$$

În cazul în care V = sp. vect. complex,

analog $\Rightarrow a_{ij} = \overline{d_{ji}} \Rightarrow D = \overline{A^T} \Rightarrow$

$$\Rightarrow \overline{(M_f^{\mathcal{B}})^T} = M_{f^*}^{\mathcal{B}}$$

Def.: Matricile A cu prop. $A = A^T$ sn. MATRICI SIMETRICE.

În rap. cu op. de înm., gr. mat. simetrice $\subset$ e un subgr. al mat. inversabile care sn. $GL(n, \mathbb{R})$.

Def.: Matricile A cu prop. că $(\bar{A})^T = A$ sn. matrici HERMITICE.

Ex.: Fie $f: \mathbb{R}^3 \to \mathbb{R}^3$, $f(x_1, x_2, x_3) = (2x_1 - x_3, x_2 + 3x_3, x_1 + x_2 - x_3)$ $\Rightarrow M_f^B =$

x_1	x_2	x_3
2	0	-1
0	1	3
1	1	-1

Cons. pe $\mathbb{R}^3$ prod. scalar $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle =$

$= x_1 y_1 + x_2 y_2 + x_3 y_3$

$f^* = ?$

Rez.:

Cons. $B = \{ e_1 = (1,0,0),\ e_2 = (0,1,0),\ e_3 = (0,0,1) \}$ bază ortonormală în sup. cu prod. scalar.

$$f(e_1) = f(1,0,0) = (2,0,1) = 2 \cdot e_1 + 0 \cdot e_2 + 1 \cdot e_3$$

$$f(e_2) = f(0,1,0) = (0,1,1) = 0 \cdot e_1 + 1 \cdot e_2 + 1 \cdot e_3$$

$$f(e_3) = f(0,0,1) = (-1,3,-1) = -e_1 + 3e_2 - e_3$$

$$\cdot\ M_f^B = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & 3 \\ 1 & 1 & -1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow M_{f^*}^{B^*} = (M_f^B)^T = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 3 & -1 \end{pmatrix} \Rightarrow$$

$$\Rightarrow \begin{cases} f^*(e_1) = 2\cdot e_1 + 0\cdot e_2 + (-1)\cdot e_3 \\ f^*(e_2) = 0\cdot e_1 + 1\cdot e_2 + 3\cdot e_3 \\ f^*(e_3) = 1\cdot e_1 + 1\cdot e_2 + (-1)\, e_3 \end{cases}$$

$$\Rightarrow f^*(x_1, x_2, x_3) = f^*(x_1 e_1 + x_2 e_2 + x_3 e_3) =$$

$$= x_1 f^*(e_1) + x_2 f^*(e_2) + x_3 f^*(e_3)$$

$$= x_1 (2e_1 + 0 \cdot e_2 + (-1) e_3) +$$

$$+ x_2 (0 \cdot e_1 + 1 \cdot e_2 + 3 \cdot e_3) +$$

$$+ x_3 (1 \cdot e_1 + 1 \cdot e_2 + (-1) e_3)$$

$= (2x_1 + 0 \cdot x_2 + 1 \cdot x_3,\ 0 \cdot x_1 + 1 \cdot x_2 + 1 \cdot x_3,\ (-1) \cdot x_1 + 3x_2 + (-1)x_3)$

Def.: Dc. V = sp. vect. finit, real sau complex

$f : V \to V$ apl. liniară ($f \in End(V)$)

$\Rightarrow f$ endomorfism

și $f = f^*$

atunci f sn. APLICAȚIE AUTOADJUNCTĂ.

Obs.: f autoadjunctă $\Leftrightarrow f = f^* \Leftrightarrow M_f^{\mathcal{B}} = M_f^{\mathcal{B}*} \Leftrightarrow$

$M_f^{\mathcal{B}} = (M_f^{\mathcal{B}})^T$ (în cazul în care $\mathcal{B}$ e bază ortonormală)

$M_f^{\mathcal{B}}$ simetrică

În cazul în care V = sp. vect. complex, finit

și $f = f^* \Leftrightarrow \overline{(M_f^{\mathcal{B}})^T} = M_f^{\mathcal{B}*}$ pt. $\mathcal{B}$ = bază

ortonormală $\Leftrightarrow \overline{(M_f^{\mathcal{B}})^T} = M_f^{\mathcal{B}}$

$M_f^{\mathcal{B}}$ hermitică.

Proprietăți ale aplicațiilor autoadjuncte

1) Dc. $f: V \to V$ liniară, $f = f^*$, V = finit, real, complex cu prod. scalar

atunci $\sigma(f) \subset \mathbb{R}$

Dem.: Fie $\lambda_0 \in \Gamma(f)$ și $v \in V_{\lambda_0}$, $v \neq 0$

$$\langle f(v), v \rangle = \langle \lambda_0 v, v \rangle = \lambda_0 \langle v, v \rangle$$

$$\| \qquad \neq 0$$

$$\langle v, f^*(v) \rangle \overset{f^* = f}{=} \langle v, f(v) \rangle = \langle v, \lambda_0 v \rangle =$$

$$= \overline{\lambda_0} \langle v, v \rangle$$

$$\neq 0$$

$$\Rightarrow (\lambda_0 - \overline{\lambda_0}) \langle v, v \rangle = 0 \Rightarrow \lambda_0 - \overline{\lambda_0} = 0 \Rightarrow$$

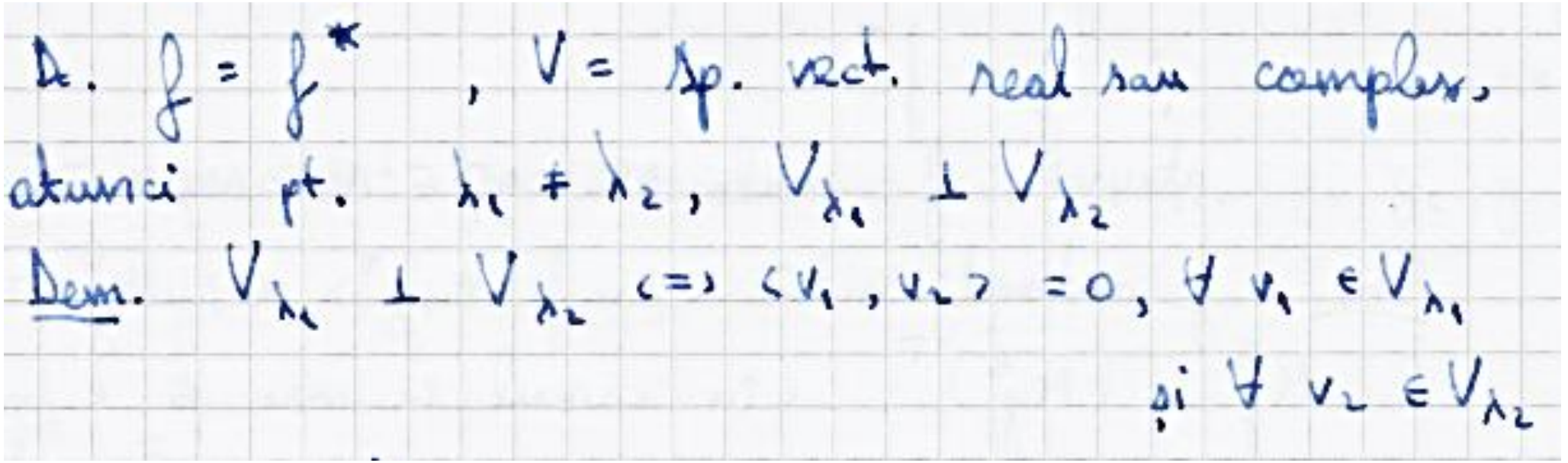
$$\neq 0$$

$$\Rightarrow \lambda_0 = \overline{\lambda_0} \Rightarrow \lambda_0 \in \mathbb{R}$$

2) În cond. de mai sus, V = sp. vect. real, $f = f^*$
$\Rightarrow T(f) \in \mathbb{R}$

Dem.: Cons. mult. scalarilor ca fiind parte din $\mathbb{C}$,
conf. prop. ant. $T(f) \subset \mathbb{R}$

3)


D. $f = f^*$, V = sp. vect. real sau complex,
atunci pt. $\lambda_1 \neq \lambda_2$, $V_{\lambda_1} \perp V_{\lambda_2}$
Dem. $V_{\lambda_1} \perp V_{\lambda_2} \Leftrightarrow \langle v_1, v_2 \rangle = 0$, $\forall v_1 \in V_{\lambda_1}$
și $\forall v_2 \in V_{\lambda_2}$

Fie $v_1 \in V_{\lambda_1}$, $v_1 \neq 0$

$v_2 \in V_{\lambda_2}$, $v_2 \neq 0$

$$\langle f(v_1), v_2 \rangle = \langle v_1, f^*(v_2) \rangle \overset{f=f^*}{=\!=} \langle v_1, f(v_2) \rangle =$$

$$= \langle v_1, \lambda_2 v_2 \rangle = \overline{\lambda_2} \langle v_1, v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle$$

$\lambda_2 = \overline{\lambda_2}$, $\lambda_2 \in \mathbb{R}$

$$\langle \lambda_1 v_1, v_2 \rangle = \lambda_1 \langle v_1, v_2 \rangle \overset{\lambda_1 \neq \lambda_2}{=\!=\!=} \langle v_1, v_2 \rangle = 0, \quad \forall v_1 \in V_{\lambda_1}, \ \forall v_2 \in V_{\lambda_2}$$

!

Temă

$$\Rightarrow V_{\lambda_1} \perp V_{\lambda_2} \iff V_{\lambda_1} \oplus V_{\lambda_2}$$

Temă:

Dc. $U, W \subset V$, $U \perp W \overset{?}{\Rightarrow} U \oplus W$

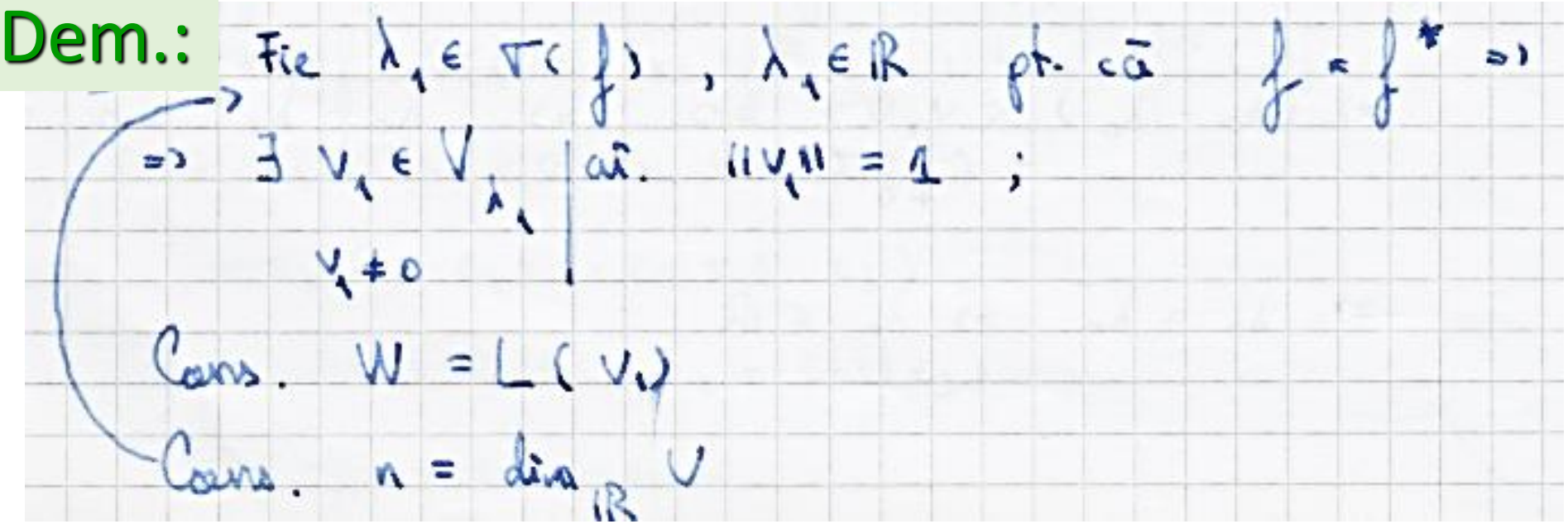
4) $f : V \to V$, V = sp. vect. real, finit, cu $\langle \cdot, \cdot \rangle$,

f liniară, $f = f^*$

atunci $\exists\, B$ bază ortonormală a lui V

a.î. M_f^B = formă diagonală

Dem.:


Fie $\lambda_1 \in \sigma(f)$, $\lambda_1 \in \mathbb{R}$ pt. că $f = f^*$ ⇒
⇒ $\exists\, v_1 \in V_{\lambda_1}$ a.î. $\|v_1\| = 1$;
$v_1 \neq 0$
Cons. $W = L(v_1)$
Cons. $n = \dim_{\mathbb{R}} V$

Cons. $W^{\perp}$ = complementul ortogonal al lui W (cu bază $B_{W^{\perp}} = \{v_1, \dots, v_n\}$ = bază ortonormată)

Dc. $v \in W^{\perp}$ atunci $\langle v_1, v \rangle = 0$

$\langle f(v), v_1 \rangle = \langle v, f^*(v_1) \rangle \overset{f=f^*}{=\!=} \langle v, f(v_1) \rangle =$

$= \langle v, \lambda_1 v_1 \rangle = \lambda_1 \underbrace{\langle v_1, v_1 \rangle}_{0} = \lambda_1 \cdot 0 = 0 \Rightarrow$

$\Rightarrow f(v) \in W^{\perp} \Rightarrow$ se poate defini $f|_{W^{\perp}} : W^{\perp} \to W^{\perp}$.

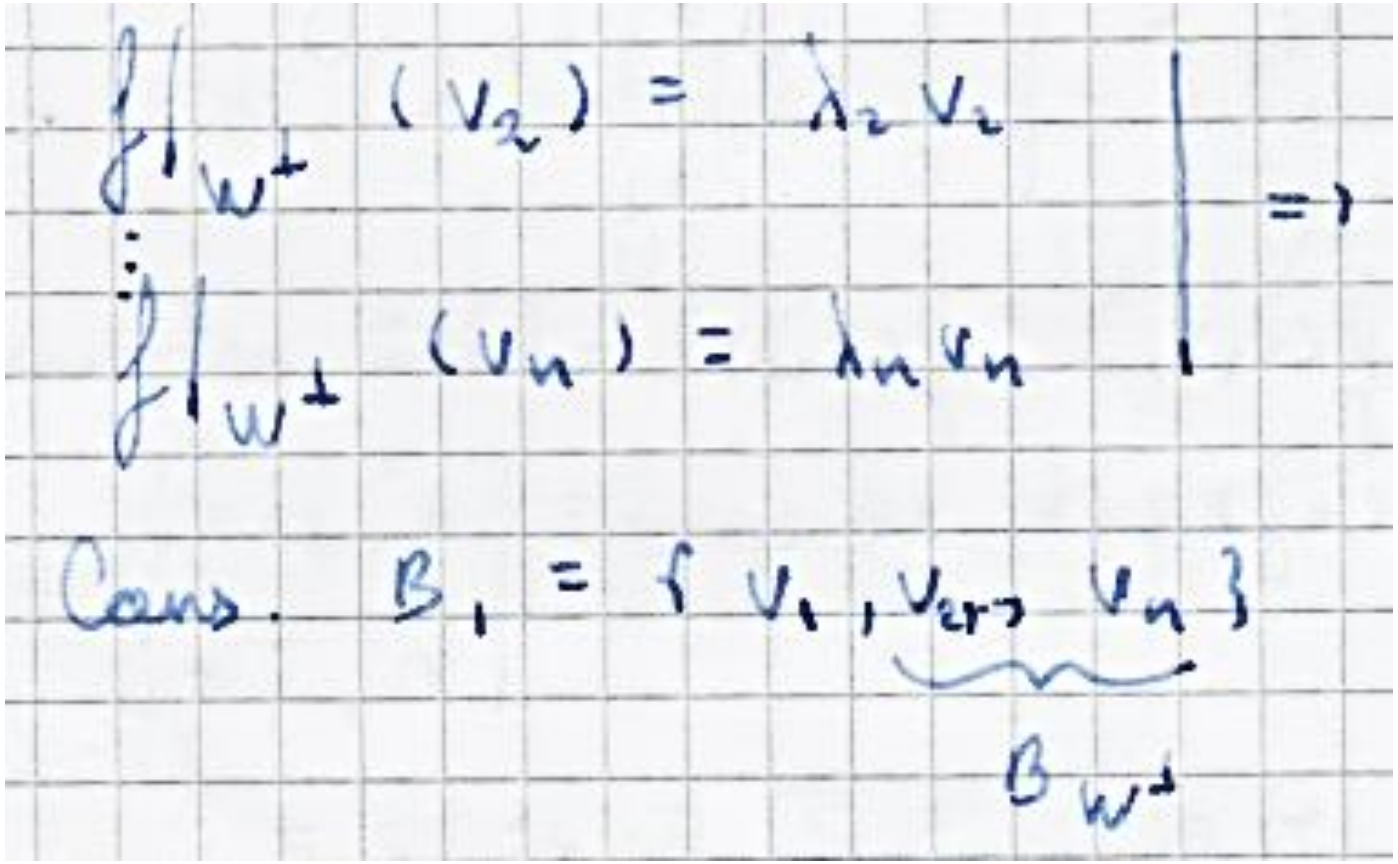
→ Dem. prop. cerută prin ind. după n

Cons. P(n-1) adev. și dem. P(n)

Cum $f = f^* \Rightarrow f|_{W^{\perp}} = f^*|_{W^{\perp}}$.

Deoarece $\dim_{\mathbb{R}} W^{\perp} = n-1 \Rightarrow \exists\ B_{W^{\perp}} = \{v_2, \dots, v_n\}$ bază ortonormală a.î. $M_{f|_{W^{\perp}}}^{B_{W^{\perp}}}$ = formă diag.

$$M_{f|_{W^{\perp}}}^{B_{W^{\perp}}} = \begin{pmatrix} \lambda_2 & \dots & 0 \\ \vdots & & \\ 0 & \dots & \lambda_n \end{pmatrix}$$


$f|_{W^\perp}(v_2) = \lambda_2 v_2$
$f|_{W^\perp}(v_n) = \lambda_n v_n$
$\Rightarrow$
Cons. $B_1 = \{v_1, v_{2r3} \underbrace{v_n}_{B_{W^\perp}}\}$

$f(v_1) = \lambda_1 v_1$

$f(v_2) = f|_{W^+}(v_2) = \lambda_2 v_2$

$\vdots$

$f(v_n) = f|_{W^+}(v_n) = \lambda_n v_n$

$\Rightarrow M_f^{B_1} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & & \\ 0 & \dots & \lambda_n \end{pmatrix}$

cu B_1 = bază ortonormală

Concluzie: Dc. o matrice e simetrică, ea poate fi cons. ca matrice a unei apl. f autoadjunctă, ca urmare matricea se poate diagonalizeza.

Transformări ortogonale

Fie $f : V \to V$, V = sp. vect. finit, real sau complex

f liniar

f e transformare ortogonală $\Leftrightarrow$

$\Leftrightarrow \forall\ B = \{e_1, e_2, \dots, e_n\}$ = bază ortonormală $\Rightarrow$

$\Rightarrow \{f(e_1), \dots, f(e_n)\}$ = bază ortonormală

Teoremă: Dc. $f : V \to V$ e liniară,

V = sp. vect. finit real

atunci sunt $\Leftrightarrow$ următoarele:

1) f e transf. ortogonală

2) f conservă prod. scalare

3) $(H_f^{\mathcal{B}})^{-1} = (H_f^{\mathcal{B}})^T$

Dem.: 1→2

Fie $B = \{e_1, \dots, e_n\}$ bază ortonormală

$$\langle e_i, e_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Fie $B = \{f(e_1), \dots, f(e_n)\}$ = bază ortonormală

$$\langle f(e_i), f(e_j) \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

$$\langle v, u \rangle = \langle \underbrace{\sum_i \alpha_i e_i}_{v}, \underbrace{\sum \beta_j e_j}_{u} \rangle =$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \beta_j \langle e_i, e_j \rangle =$$

$$= \sum_{i=1}^{n} \alpha_i \beta_i$$

$$\langle f(u), f(u) \rangle = \langle f(\Sigma \alpha_i e_i), f(\Sigma \beta_j e_j) \rangle =$$

$$= \langle \sum_{i=1}^{n} \alpha_i f(e_i), \sum_{i=1}^{n} \beta_j f(e_j) \rangle =$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \beta_j \underbrace{\langle f(e_i), f(e_j) \rangle}_{\delta_{ij}} =$$

$$= \sum_{i=1}^{n} \alpha_i \beta_i$$

$\langle v, u \rangle = \langle f(v), f(u) \rangle \Rightarrow f$ conservă prod. scalare

Obs. O transf. ortogonală păstrează lungimea (norma) vectorilor și unghiurile dintre vectori.

2→3

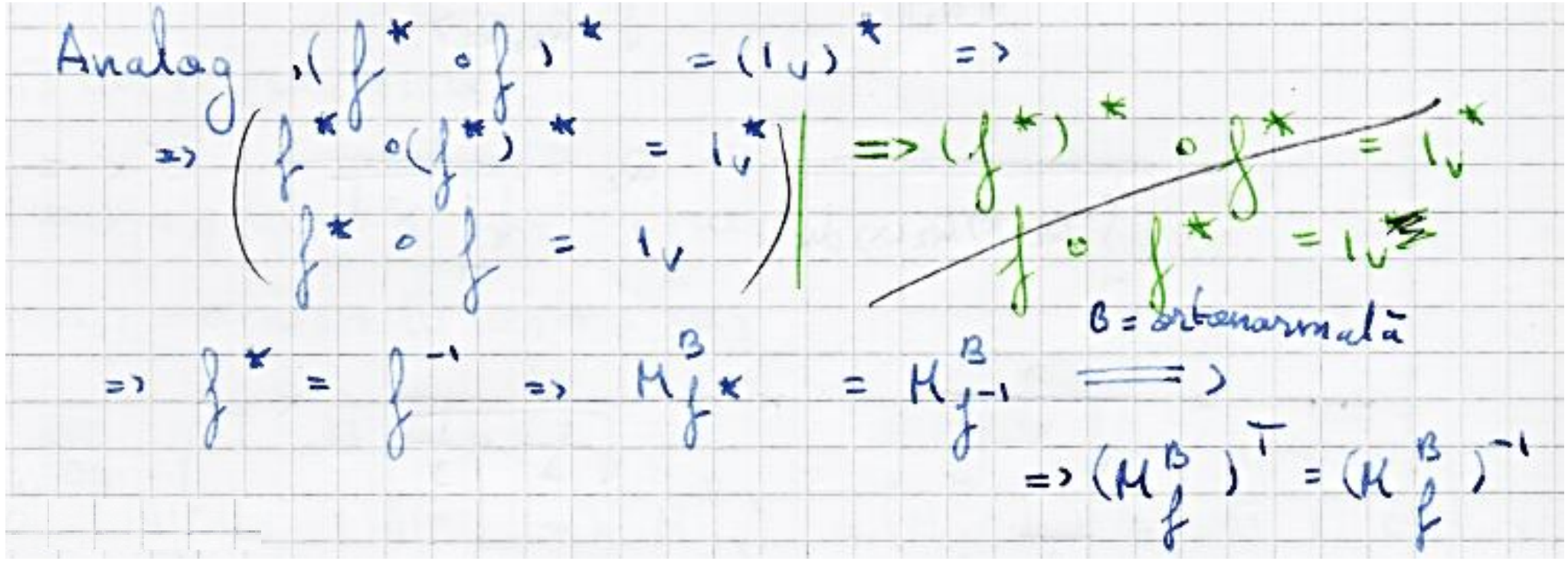
$\langle f(v), f(u)\rangle = \langle v, u\rangle$, $\forall\, v, u \in V$

$\langle v, f^* \circ f(u)\rangle = \langle v, 1_V(u)\rangle$, $\forall\, v, u \in V$

$1_V : V \to V$

$1_V(v) = v$

$\Rightarrow f^* \circ f = 1_V$


Analog $(f^* \circ f)^* = (1_V)^*$ =>
=> $f^* \circ (f^*)^* = 1_V^*$
$f^* \circ f = 1_V$
=> $(f^*)^* \circ f^* = 1_V^*$
$f \circ f^* = 1_V^*$
B = ortonormală
=> $f^* = f^{-1}$ => $M_{f^*}^B = M_{f^{-1}}^B$ ==>
=> $(M_f^B)^T = (M_f^B)^{-1}$

3→1

$\langle f(e_i), f(e_j)\rangle = \langle e_i, f^*(f(e_j))\rangle =$

$= \langle e_i, f^{-1}(f(e_j))\rangle = \langle e_i, e_j\rangle \Rightarrow f$ tr. ortogonală